Statement of the problem of creating an application for finding a sea vessel's course

Aim

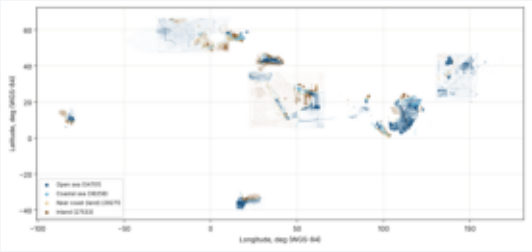
Develop a cross-platform desktop application that searches for the optimal course of a sea vessel over open-sea points with a genetic algorithm and compares it with Dijkstra and A*.

Input: labelled dataset (WGS-84 points, 4 classes), departure and destination, parameters of the graph, penalties for dangerous zones and parameters of the genetic algorithm.
Output: route of the GA, optimal routes of Dijkstra and A*, convergence plot, comparison table, files (PNG, GeoJSON, CSV, parameter log).

Dataset of the student (after cleaning)

Class	Meaning	Points	Role
OPEN_SEA	open sea	54 701	node
COASTAL_SEA	coastal (shallow) sea	36 258	node, penalised
NEAR_COAST	land within 5 km of shore	29 271	obstacle
COASTLINE	inland	27 533	obstacle

152 838 points read 5 075 removed
(class contradicts the measured depth, tolerance 5 m)



Requirements of the assignment

– Windows 10/11 and Linux (Ubuntu 20.04+, Fedora 38+); graphical user interface;
– loading of the dataset from a file (WGS-84, 4 classes of points);
– graph over the points: by distance or by visibility;
– genetic algorithm for the ship's course; avoidance of dangerous zones, minimisation of the path length;
– settings: population size, generations, crossover and mutation probabilities, penalties for dangerous zones;
– Dijkstra and A* for comparison on the same graph;
– map with the points and the three routes, convergence plot of the GA;
– export: map image, routes, convergence plot, parameter log.

Criterion: route cost $F(P) = L(P) + \alpha \cdot Lc(P) + \beta \cdot Ld(P) \rightarrow \min$
L – length, Lc – length in coastal water, Ld – danger-weighted length near land

Rev.	Sheet	Doc. No.	Sign.	Date
Developed		Li H.		
Checked		Kozinsky A.A.		
T. contr.				
N. contr.				
Approved		Golovko V.A.		

КР.ИИ2025.250978-01

Development of an application for finding the optimal course of a sea vessel over open-sea points based on a genetic algorithm

Problem statement (poster)

Lit.	Mass	Scale
	–	–

Sheet 1 | Sheets 5

BrSTU, Dept. of IIT
group ИИ-2025